

A SERS investigation of the electrodeposition of Ag–Au alloys from free-cyanide solutions

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Abstract

In this paper we investigated the electrodeposition of Ag–Au alloys from cyanide baths by in situ surface-enhanced Raman spectroscopy. The spectroelectrochemical investigation is complemented by electrochemical experiments (cyclic voltammetry and potentiostatic electrodeposition) and structural (X-ray diffraction), morphological (scanning electron microscopy) and compositional (energy dispersive spectroscopy) characterisation of the alloys. Electrochemical and spectroelectrochemical results on single-metal electrodeposition baths of otherwise identical composition are reported for comparison. Single-phase deposits of composition ranging from 13% to 75% Au were obtained potentiostatically. The voltammetric behaviour was correlated to the vibrational characteristics of adsorbed ligands. Quantitative and qualitative differences in interfacial chemistry as a function of applied potential were assessed and explained on the basis of variations in the nature of cyanide-related surface species. The cyanide and metal-adsorbate stretching bands have been analysed.

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1. Introduction

Electrodeposition of gold and silver from free-cyanide baths has been common industrial practice since the early times of electrochemical plating and electroforming towards the end of the 19th century. Some detailed reviews on this topic have been published [1,2]. Extensive operative knowledge of the behaviour of these systems has accumulated during the decades and these baths are universally recognised as state-of-the-art reliable technologies. Electrochemical investigations on this topic have mainly concerned the definition of the mechanism of metal deposition from their cyanocomplexes [3–7]. Recently, the research has been mainly directed towards the definition of alternative cyanide-free baths [8] and several patents have been filed, chiefly regarding the use of sulphite, thiourea, thiosulphate or

other sulphorated compounds as metal ligands; a comprehensive list of the patent literature would be well beyond the scope of this paper. A more limited range of literature sources is available on the electrodeposition of gold–silver alloys, essentially restricted to the patent literature.

In the definition of deposition mechanisms and metal reduction kinetics and in the perspective of technological improvements, a key role is played by adsorption phenomena at the interface, which may involve both electroactive and non-electroactive species such as ligands, organic additives and supporting electrolyte ions. The radiotracer method has been applied to the study of the adsorption of cyanide-containing species from a potassium dicyanoaurate alkaline solution, in a wide potential range, including the electrodeposition region. The results have been discussed in comparison with those obtained from a potassium cyanide solution [9]. Notwithstanding the valuable results obtained, this technique being an integral one, it is not ideal in providing information on

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the chemical identity of the adsorbed species. This purpose can be achieved by the use of spectroscopic methods, able to identify the chemical nature and the orientation of species at the metal|electrolyte interface. In the last few decades the application of in situ electrochemical vibrational spectroscopies [10,11] has allowed a deeper understanding of the electrode processes by directly monitoring the adsorption phenomena. Several in situ SERS studies of cyanide adsorbed on gold and on silver are available in the literature [12–16]. The situation is similar for in situ FT-IR [17–19]. A smaller amount of spectroelectrochemical work has been carried out during the electrodeposition of silver [20–26] and gold [26–28].

Gold and silver are well known SERS-active metals. What the SERS activity is actually due to, has not been ascertained unambiguously, so far. A giant (order of 10^6) enhancement of the Raman signal was observed from molecules adsorbed on adequately roughened metals and this enhancement effect was clearly seen to be dependent on the nature of the metal, so that a distinction and classification among coinage metals, alkali metals and transition metals could be made [11]. The differences in enhancement effects across these classes of metals can be traced to the respective plasmon resonance frequencies. A given degree of roughness is needed in order to produce a suitable surface geometry giving rise to optimal interaction between the electric field of the incident light and the surface plasmons [29]. Roughening is usually accomplished by an electrochemical procedure consisting of subsequent oxidation and reduction cycles. Alternatively, a suitable roughness can develop spontaneously during electrocrystallisation [22,27]. Recently, interest in the SERS behaviour of alloy electrodes has arisen [30–34], with an eye to understanding the effects caused by mixing metals with different SERS activity. A preliminary report of SERS work with Ag–Au colloidal metal films of various compositions has been published [35]. FT-IR based spectroelectrochemical work on the electrodeposition of Ag–Au has recently been published [26,36]. At variance with this paper, in which reflectance FT-IR has been used to observe the vibrations of bulk species present in the thin electrolyte layer adjacent to the working electrode, the use of confocal surface-enhanced Raman spectroscopy in our work provides further enhancement of the surface sensitivity to the surface species at the electrode|solution interface during electroplating. In our laboratory, we have lately carried out similar SERS experiments during the electrodeposition of Au–Sn alloys [33,34]. In the present paper, we present SERS data acquired during the electrodeposition of Ag–Au alloys, from a free-cyanide bath, and, for comparison, results obtained under the same experimental conditions from the analogous gold and silver baths.

2. Experimental

2.1. Materials

The following aqueous solutions were employed: (i) 50 mM $\text{KAu}(\text{CN})_2$, 0.2 M KCN, 0.5 M KOH; (ii) 50 mM $\text{K}_2\text{Ag}(\text{CN})_3$, 0.2 M KCN, 0.5 M KOH; (iii) 50 mM $\text{KAu}(\text{CN})_2$, 50 mM $\text{K}_2\text{Ag}(\text{CN})_3$, 0.2 M KCN, 0.5 M KOH. Analytic grade chemicals were used. The solutions were prepared with ultra-pure water (resistivity 18 $\text{M}\Omega\text{ cm}$) obtained with a Milli-Q system. The baths were operated at room temperature (ca. 24 °C).

2.2. Cyclic voltammetry

Cyclic voltammetry (CV) was carried out in a standard three-electrode cell. An AMEL 5000 potentiostat was used. The CVs for the Ag bath were measured with an Ag electrode, those corresponding to the Au and Ag–Au baths with an Au electrode. The working electrodes were polycrystalline wires (diameter 1 mm, immersed length 20 mm) mounted in an AMEL 493 electrode holder. The reference electrode was $\text{Ag}|\text{AgCl}$ 13 M KCl in all experiments, including spectroelectrochemical ones. Potential values are reported vs. $\text{Ag}|\text{AgCl}$. The reference electrode tip was placed at a distance of ca. 5 mm from the working electrode. Ohmic corrections applied by the positive feedback method were considered, but proved irrelevant. The counter electrode was a platinum foil (area 3 cm^2). Unless otherwise stated the scan rate was 20 mV s^{-1} , the initial scan direction was from positive to negative.

2.3. X-ray diffractometry, scanning electron microscopy and energy dispersive spectroscopy

Structural characterisation was performed by X-ray diffraction (XRD), using a Philips PW 1830 diffractometer equipped with a Philips PW 1820 vertical powder goniometer and a Philips 1710 control unit. The scan rate was 1 deg s^{-1} . The radiation was unmonochromated $\text{Cu K}\alpha$. The morphology of the samples was studied using a Cambridge Stereoscan 360 SEM. The electron source was LaB_6 . Electron detection was carried out with a scintillation photodetector. The typical working pressure was 10^{-7} mbar. Quantitative analysis of the sample composition was performed by EDX in the same vacuum chamber and with the same electron source as SEM, with a Li-doped Si detector. The samples for compositional and structural investigations were electrodeposited potentiostatically at -700 , -900 , -1100 and -1300 mV (thicknesses in the range 10–35 μm) onto ca. 30 μm thick amorphous electroless Ni–P (P 9%) layers prepared according to a procedure described elsewhere [37]. The working electrode active area was 4 cm^2 . A prismatic cell holding 300 ml of plating solution

was employed. A plane-parallel electrode configuration was adopted. The reference electrode was connected to the catholyte through a Piontelli probe [38]. The tip of the Piontelli probe was located at a distance of ca. 30 μm from the cathode surface, thus minimising the ohmic drop.

2.4. *In situ* Raman spectroscopy

SERS measurements were carried out with a Raman microprobe system (LabRam Jobin-Yvon) equipped with a true confocal microscope, CCD detector and holographic notch filter. Excitation at 633 nm was provided by a He–Ne laser and the power delivered at the sample was about 12 mW. A 10 $\times$ long-working-distance objective was used. *In situ* electrochemical measurements were performed in a Ventacon cell with a vertical polycrystalline Au disc working electrode of diameter 3 mm embedded into a Teflon holder. The working electrode was located at a distance of ca. 3 mm from the optical window, thus ensuring suitable current density distribution and mass transport conditions for electrodeposition during spectroelectrochemical measurements. Natural convection at low plating rates and forced convection due to hydrogen evolution at higher negative potentials guarantee adequate electrolyte homogeneity. The electrolyte depletion during SERS experiments was estimated to be lower than 10%, which ensures the compositional consistency of the electrodeposited alloys and lack of serial correlation among potentiostatic experiments. Metallographic polishing with 1200 grit paper before each experimental run proved sufficient for excellent reproducibility of spectroelectrochemical measurements. The counter electrode was a Pt wire loop (1.25 cm²) concentric with the working electrode disk. The acquisition time was varied in the range 10–500 s in order to maximise the signal-to-noise ratio, typical values are around 250 s but significantly lower figures apply at intermediate negative voltages.

The working electrodes were not roughened with the traditional *orc* procedures, but SERS activity was achieved via the electrocrystallisation process, which produces SERS-active sites during electrodeposition. Surface enhancement was noticed immediately after the electrodeposition reaction had started; the enhancement factor increased with the plating rate and it eventually decreased in the hydrogen evolution potential range, owing probably to radiation scattering by bubbles.

3. Results and discussion

3.1. Cyclic voltammetry

The electrochemical behaviour of the three relevant systems has been investigated by means of CV in the

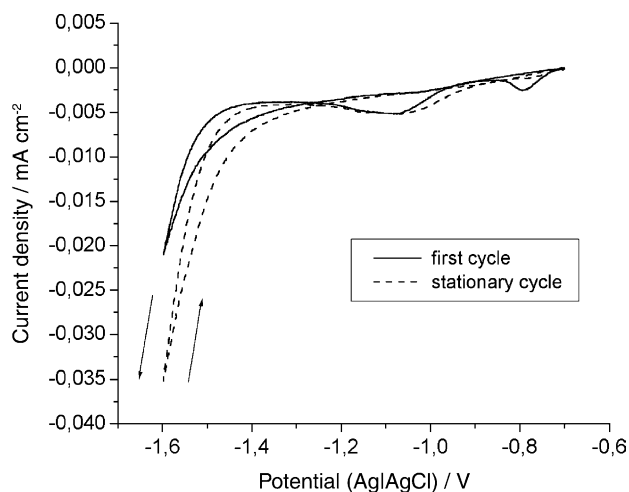


Fig. 1. Cyclic voltammogram of an Au electrode in contact with the Au electrodeposition bath. Scan rate: 20 mV s⁻¹.

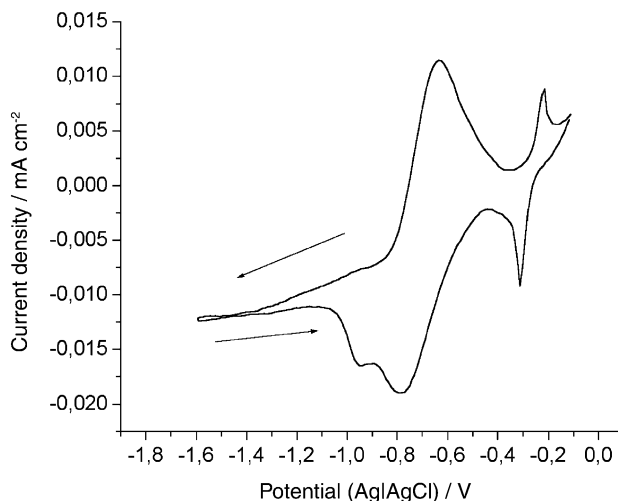


Fig. 2. Cyclic voltammogram of an Ag electrode in contact with the Ag electrodeposition bath. Scan rate: 20 mV s⁻¹.

potential range spanned by the Raman experiments. The respective graphs are reported in Figs. 1–3.

3.1.1. Gold electrodeposition bath

The curves referring to Au (Fig. 1) show two irreversible reduction peaks. The one at lower negative potential (–750 mV) is prominent in the first cycle and its intensity tends to decrease with subsequent cycling. As expected, this feature can be enhanced if higher scan rates, e.g., 100 mV s⁻¹, are employed. This peak can be related to the peculiar inhibited Au deposition mechanism active at low current densities [39], where a peculiar crystallographic structure was assessed and explained with the intercalation of CN⁻-related species in the crystal lattice. In the literature a chemical–electrochemical mechanism with an adsorbed AuCN species

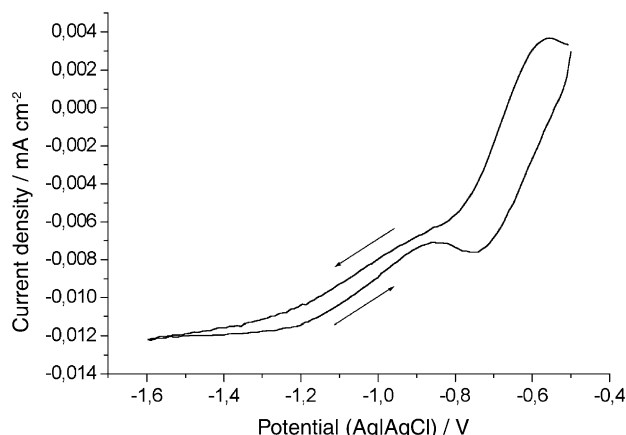


Fig. 3. Cyclic voltammogram of an Au electrode in contact with the Ag–Au electrodeposition bath. Scan rate: 20 mV s^{-1} .

has been proposed for the reduction of the Au(I) cyanocomplex at low negative potentials [3,4,40]. At high negative polarisations, Au electrodeposition has been reported to take place according to a different mechanism, which can be interpreted as the direct reduction of $\text{Au}(\text{CN})_2^-$ to metallic Au [40]. Also this kind of mechanism gives rise to negative passivation starting around -1000 mV in accord with thermodynamic data [41]. Hydrogen evolution starts around -1400 mV and the respective partial current density is added to that corresponding to Au reduction. The lower current density in the reverse negative scan seems to be typical for an interface modified by Au deposition (high surface coverage with adsorbed CN^-), on which water reduction occurs with a local increase of pH [41]. A modification of the passivating cyanide layer has been noticed at high negative polarisations, within the hydrogen evolution range, giving rise to an enhancement of the electrodeposition rate. Once these depolarised electrodeposition conditions have been attained during CV measurements, the electrode is modified and if subsequent voltammetric scans are imposed, higher terminal negative current densities are obtained, in addition to the variation of the peak intensities noted above.

3.1.2. Silver electrodeposition bath

The voltammetric behaviour of the Ag bath at an Ag electrode is shown in Fig. 2. The scan dependence of the voltammetric features is much less pronounced in the Ag than in the similar Au bath. A steady-state voltammogram is reported in Fig. 2. The first scan (not shown, for the sake of clarity) exhibits only some minor quantitative differences in the peak intensity, probably relating to the achievement of a steady-state electrode surface morphology. In the more positive range (ca. -200 mV) the reversible formation and reduction of Ag oxide can be noticed. We are not interested in defining the exact nature of this compound, which is most probably Ag_2O

[42], even if more recent investigations in alkaline solutions highlight the formation of several hydroxide and hydrous oxide species [43]. The voltammetric features present in the negative-going scan are similar to those observed with the Au bath: a sequence of two negative peaks can be noticed at ca. -800 and -1000 mV , respectively, probably corresponding to chemical–electrochemical and electrochemical reduction mechanisms. The hydrogen evolution reaction cannot be noticed within the potential range investigated, reasonably owing to a higher hydrogen overvoltage in this system as compared to the analogous Au bath [44]. This point deserves a more in-depth discussion and is expected to have a significant bearing on Ag electrodeposition [45], but we consider that this would be beyond the scope of the present paper. At variance with the Au bath, the Ag system exhibits an oxidation peak at potentials more negative than those required for the formation of Ag oxide. This peak can be explained with the formation of $\text{Ag}(\text{CN})_2^-$ by corrosion of Ag in the free- CN^- solution [3,46]. From these data, it seems apparent that Ag is more noble than Au in a cyanide electrolyte, as expected from the relative order of stability of the cyanocomplexes of the two metals.

3.1.3. Alloy electrodeposition bath

When both metal ions are present in the electrodeposition bath, Au is deposited simultaneously with Ag within a wide range of electrochemical conditions also encompassing very low current densities [26,47] (see Section 3.2). As the current density is increased, the Au content in the deposit increases as well, steeply at the beginning, then more slowly. An asymptotic voltammogram for the alloy system is shown in Fig. 3. A very limited hysteresis of the voltammetric behaviour is observed with the alloy bath. Also in this case two peaks are noticed in the negative-going scan. They are less defined in the alloy solution than in the single-metal baths. The position of the first, better defined peak (ca. -700 mV) is approximately the same as for Ag and Au. The second feature is hardly visible and it tends to reach an asymptote at ca. -1200 mV , a potential similar to the peak position of the second feature of the Au system. The negative current density recorded in the voltammogram of Fig. 3 is thus believed to be due to the simultaneous discharge of Ag and Au ions. It is less straightforward to assign specific reduction mechanisms to the voltammetric features in the alloy bath; nevertheless the less negative peak can be thought of as related to a chemical–electrochemical mechanism, while the more negative limiting current density might imply that direct discharge of the cyanocomplex becomes involved. Also in the alloy bath a high hydrogen overvoltage is noticed. The behaviour of the positive-going scan is somehow intermediate between that of Ag and Au solutions. A very faint positive feature is noticed in the

higher negative range, where the Au content in the deposit is higher, while a positive peak appears at less negative values, corresponding to Ag-enriched deposits. This peak is possibly due to the formation of cyano-complexes by alloy corrosion in the CN^- environment.

3.2. Composition, structure and morphology of potentiostatically grown layers

3.2.1. Alloy composition

The composition of the alloys, as evaluated by EDX, is reported in Fig. 4. The error bars refer to three replicated EDX measurements (scanned area ca. $50 \times 50 \mu\text{m}^2$) carried out in different positions of the same electrodeposits. Compositional and morphological (see Section 3.2.3) edge effects were verified to be minor ones (hardly measurable). The expected regular codeposition behaviour was verified, the slight differences in the concentration of the metal salts can explain the quan-

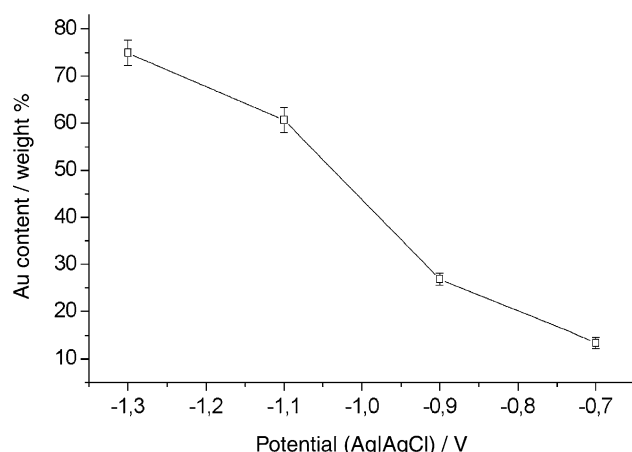


Fig. 4. Composition of Ag–Au electrodeposited alloys, as measured by energy dispersive spectroscopy.

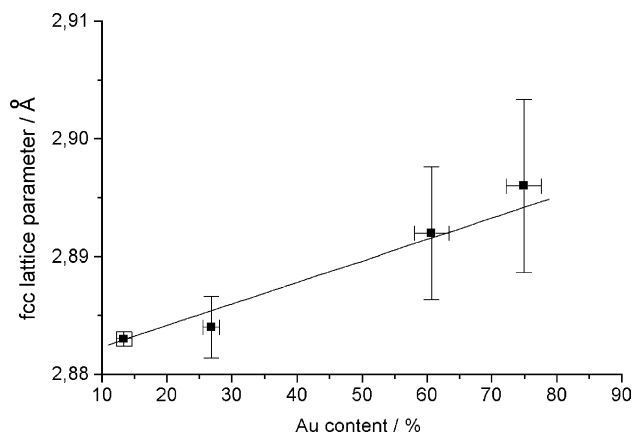


Fig. 5. Relationship between lattice parameter of the fcc electrodeposited Ag–Au alloys, derived from X-ray diffraction, and composition, derived by energy-dispersive spectroscopy.

titative differences between our results and those reported in [26,36]. The alloy composition has also been correlated with the lattice parameters derived from all the reflections of the X-ray diffractograms discussed in Section 3.2.2, according to Végard's law [48]; the expected linear correlation between lattice parameters and alloy composition has been assessed (Fig. 5).

3.2.2. Crystalline structure of the alloys

As far as the crystallographic structure of the deposits is concerned, single-phase fcc layers were grown at all the applied potentials, the respective X-ray diffractograms are shown in Fig. 6. The potentiostatic electrodeposits exhibit a slight (111) texture. The relative intensity of the (220) reflection tends to increase at higher negative potentials. The coherently diffracting domains are nanometric and their dimensions tend to decrease with increasing negative overvoltage. $K\alpha_2$ -corrected estimates of crystallographic grain size obtained by the Scherrer method [49] indicate sizes in the range 40–80 nm, positively correlated with electroodic potential. The development of electrodeposits of fcc metals with (111) faces normal to the growth direction is well documented. In particular, since the crystal facets with lower energy tend to be parallel to the deposit surface under highly inhibited conditions, the presence of the (111) orientation is compatible with higher-inhibition conditions. We recall that, for fcc crystals, the surface energy varies in the order $(220) < (200) < (111)$ [50]. The increase of the (220) orientation is thus typical of lower-inhibition conditions, this is coherent with our voltammetric and SERS results (see Sections 3.1 and 3.3, respectively).

3.2.3. Surface morphology of the alloys

The potentiostatic growth procedure for the relevant samples has been described in Section 2.3. The in-plane morphology of potentiostatic deposits is shown in Figs. 7–10. The alloy grown at -700 mV exhibits a compact

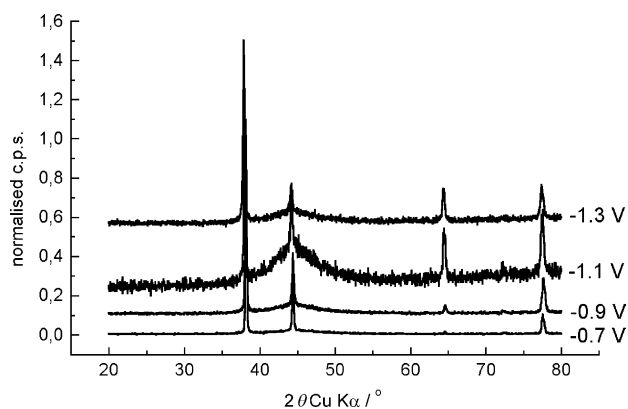


Fig. 6. X-ray diffractograms of potentiostatically deposited Ag–Au alloys.

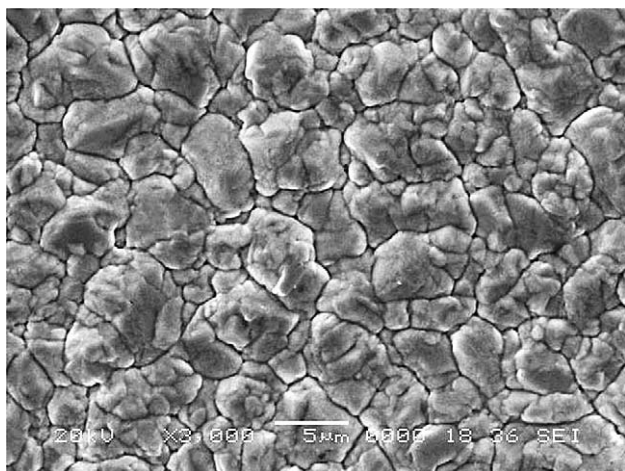


Fig. 7. SEM micrograph of Ag–Au (Au 13%) alloy electrodeposited at -700 mV vs. Ag|AgCl, magnification $3000\times$.

morphology with a wide size distribution of polygonal crystals (Fig. 7), the deposits obtained at -900 (Fig. 8) and -1100 mV (Fig. 9) display globular and cauliflower protrusions, respectively, and the layers produced at -1300 mV (Fig. 10) show an amplification of the electrode coarseness accompanied by the formation of dendrites.

The surface coarseness for a given quantity of potentiostatically electrodeposited metal under mixed-control conditions is known to be a steeply growing function of the current density [51]. The compact morphology of the alloy grown at -700 mV, exhibiting crystallites several micrometers across, are typical of the activation-controlled electrodeposition of fcc metals [52]. This interpretation is compatible with the voltammetric behaviour observed in Fig. 3. If some mass transport contribution to electrokinetic control is present, the formation of large, well-defined crystallites is no

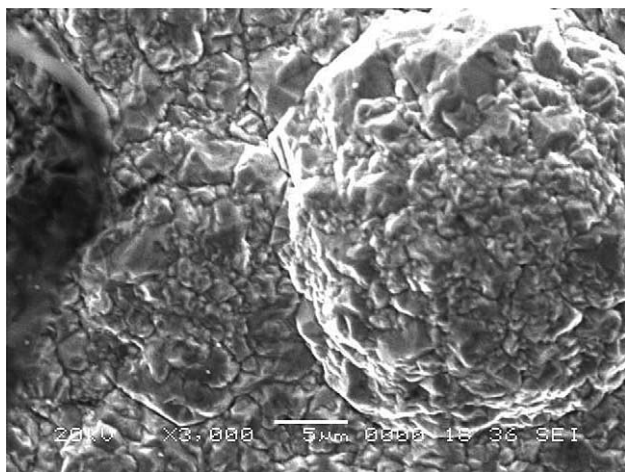


Fig. 8. SEM micrograph of Ag–Au (Au 27%) alloy electrodeposited at -900 mV vs. Ag|AgCl, magnification $3000\times$.



Fig. 9. SEM micrograph of Ag–Au (Au 61%) alloy electrodeposited at -1100 mV vs. Ag|AgCl, magnification $1000\times$.



Fig. 10. SEM micrograph of Ag–Au (Au 75%) alloy electrodeposited at -1300 mV vs. Ag|AgCl, magnification $1000\times$.

longer to be expected, because of the onset of irregularity-amplification mechanisms [53]. The morphology of electrodeposits obtained at potentials of -900 mV and lower, together with the CV features of the alloy bath, are compatible with electrocrystallisation processes under mixed control. The cauliflower growth form observed at -1100 mV (Fig. 9) has been explained by the formation of spherical diffusion conditions around the whole surface of a protrusion, due to the increase of the height-to-radius ratio of a protrusion. Under these conditions second-generation protrusions can grow inside the diffusion layer of first-generation ones in the same way as first-generation ones do within the diffusion layer of the macroscopic electrode [54]. At -1300 mV outgrowth features can be observed (Fig. 10), many of them act as dendrite precursors. Dendrites with typical branching morphologies can be detected (Fig. 10). Dendritic growth has been extensively

described in the electrocrystallisation literature (see e.g. [55]) and is mainly related to activation control or higher-dimensional diffusion contributions at the protrusion tip with diffusion control prevailing at the macroelectrode. This outgrowth morphology actually exhibits activation-control features at the tip (relatively large, well-defined crystallites) at their initial growth stages, while spherical mass transport controlled features (poorly defined crystallites and globular sub-morphologies) appear at later growth stages. The growth conditions achieved at -1300 mV are probably within the potential interval limited by the critical potentials for incipient and fully developed dendritic growth, respectively.

3.3. In situ surface-enhanced Raman spectroscopy

3.3.1. CN^- stretching bands

The SERS spectra have been recorded in the electrochemical cell varying the potential from the open circuit value (ocp) in the negative direction to -1600 mV in steps of 100 mV. The spectra were measured after having achieved a steady state of the current, which typically required ca. 5 min. We have investigated the spectral region encompassing the CN triple bond and the M–X ($X = \text{C}, \text{N}$) stretching. Three replicate series of experiments have been performed in order to achieve a sound level of statistical reliability in view of a quantitative analysis of the relevant spectral features. A remarkable degree of reproducibility has been assessed. Typical sets of potential-dependent spectra for the systems Au, Ag and Ag–Au are reported in Figs. 11–13, respectively. For the sake of compactness of presenta-

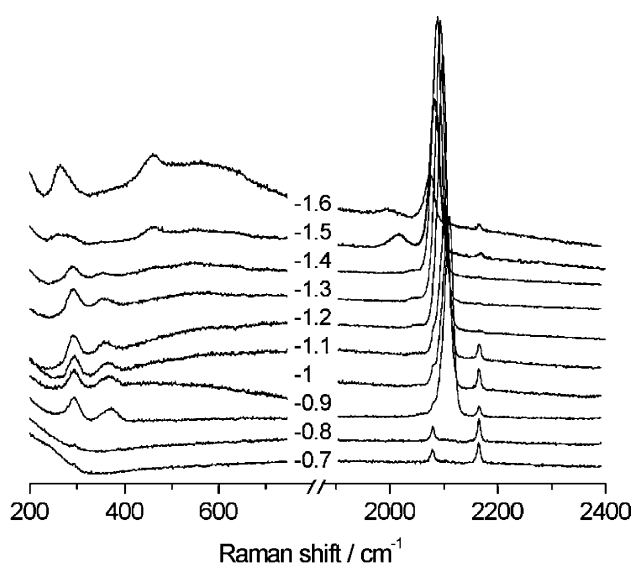


Fig. 11. In situ SERS spectra measured in the Au electrodeposition bath at various electrodic potentials. The spectra have been acquired sequentially by stepping the potential from -0.7 to -1.6 V vs. Ag|AgCl.

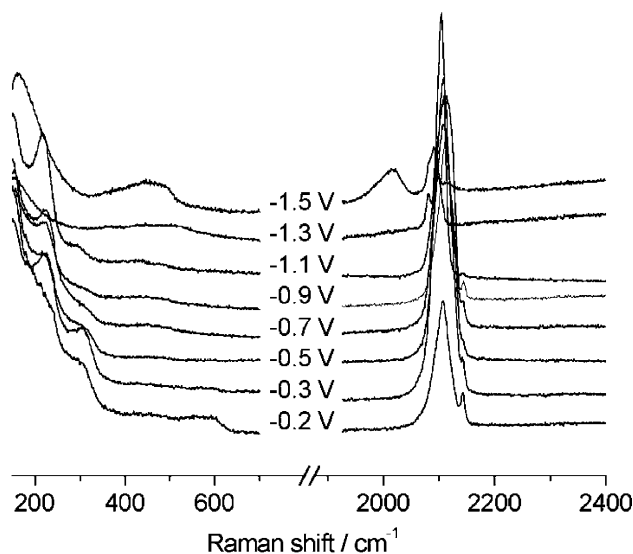


Fig. 12. In situ SERS spectra measured in the Ag electrodeposition bath at various electrodic potentials. The spectra have been acquired sequentially by stepping the potential from -0.7 to -1.6 V vs. Ag|AgCl.

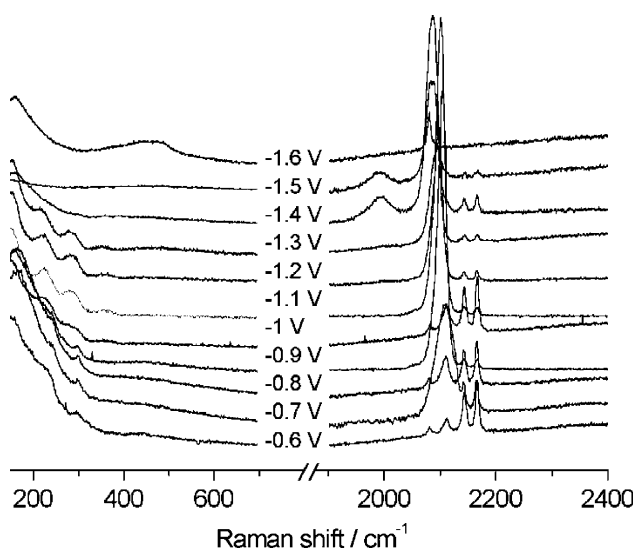


Fig. 13. In situ SERS spectra measured in the Ag–Au electrodeposition bath at various electrodic potentials. The spectra have been acquired sequentially by stepping the potential from -0.7 to -1.6 V vs. Ag|AgCl.

tion, only one out of the three sets of replicate spectra is shown for each system, since only small quantitative differences emerge among the replicates. The three electrodeposition systems, Au, Ag and Ag–Au will be analysed separately first. A comparative treatment will then follow.

In the three systems the peak due to surface CN^- (ca. 2100 cm^{-1}) appears when electrodeposition starts. The corresponding signal has the high intensity typical of a SERS band. The surface peak is not the only CN^-

stretching band appearing in the spectrum: features from the metal complex cyanide and from free cyanide in solution are easily recognisable. $\text{Au}(\text{CN})_2^-$ and $\text{Ag}(\text{CN})_2^-$ reside in the bulk of the solution. Nevertheless, provided limiting current density conditions are not achieved during electrodeposition, a finite surface concentration of these electroactive species is expected to be located at the surface, even though the actual physical nature of this proximity has not yet been fully ascertained. These cyanocomplexes can be actually adsorbed at the electrode or simply reside within the double layer. The situation is clearer for CN^- , since the band positions of adsorbed and solution species has been evaluated very accurately by IR measurements (see e.g. [56]): the band at 2080 cm^{-1} must be assigned to solution species. Nevertheless this band can be regarded as undergoing surface enhancement. As a matter of fact, from the true confocal optical configuration of the spectrometer used in this study, we can confidently exclude spectral contributions from the bulk of the solution. A reasonable explanation for such an enhancement can be provided also within the simple framework of the electrostatic enhancement for spherical features. The enhancement factor G scales with the distance r from the surface of a spherical metal particle of radius a as $G(r; a) \sim [a/(a+r)]^{12}$ (see e.g. [57]). If r ranges within values typical for the double-layer thickness ($\sim 0.5\text{ nm}$ [58,59]) and typical surface features are considered ($10\text{--}100\text{ nm}$) some degree of surface enhancement can still be anticipated if the ion is present in the double layer even though it is not directly adsorbed on the metal surface. This feature has not been observed in previous investigations employing baths not containing free cyanide [27,28].

On gold (ocp around -600 to -700 mV , Fig. 11), a surface feature in the CN^- stretching range appears for potentials of -900 mV and more negative, with a peak position ranging from 2110 to 2075 cm^{-1} ca. The shift of the peak position with potential for surface species (Stark-tuning) can be related to interfacial phenomena relevant to electrocrystallisation. A well-defined band due to the CN^- stretching in the $\text{Au}(\text{I})$ complex is found at 2165 cm^{-1} in the potential range between -700 mV and -1100 mV . This band disappears in the range -1200 to -1400 mV , where electrodeposition occurs at the limiting current density (vanishing surface concentration of electroactive species). A faint peak at this position can be observed at -1500 and -1600 mV , where intensive hydrogen evolution occurs. The increased agitation due to bubbles detaching from the cathode might give rise to a decrease of the hydrodynamic double layer thickness with enhanced mass transport and increase of the limiting current density value: this might lead to the fact that the surface concentration of $\text{Au}(\text{CN})_2^-$ is again finite. The disappearance of this band at the potential threshold of -1200 mV seems to correspond to the start

of unstable cathode growth, as confirmed by the current density, which increases progressively, at fixed potential, with elapsing time, without reaching a stationary value. The SEM micrographs reported in Section 3.2 confirm this view. The correlation between these two pieces of evidence can in fact point to a change in the kinetics and probably in the mechanism of deposition of the complex. This band does not show a measurable shift with potential, providing evidence that the relevant $\text{C}\equiv\text{N}$ -species is not adsorbed at the cathode. The free- CN^- stretching band appears around 2080 cm^{-1} . This band is fairly discernible from the surface one, because of the markedly different position (ca. 30 cm^{-1}) at low negative potentials. As the potential increases negatively, the free- CN^- band shows up as a shoulder in the intense surface- CN^- peak, which shifts progressively to lower wavenumbers owing to the Stark-tuning. The strong surface CN^- stretching peaks can actually consist of several components, as suggested by the asymmetrical shape of the peak [27]. We recall that the generally accepted theoretical shape of a SERS peak is lorentzian; in the case of CN^- stretching, a FWHM of about 30 cm^{-1} is typically reported [11,60]. The deconvolution and quantitative analysis of the different components of the experimental peak is beyond the purposes of this work.

The ocp recorded in the case of silver is much lower than that found on gold and is located around 0 mV , coherently with literature data [61]. The spectrum recorded at -100 mV displays a single-vibrational feature at 2143 cm^{-1} due to the silver cyanocomplex. The bath chemistry suggests that this species is $\text{Ag}(\text{CN})_2^-$. At variance with [26,36], where different silver complexes are seen at the interface, we observe a single species. A reason for this difference can be found in the different mass transport behaviour occurring in the experimental set-up of the two techniques: while in FT-IR, the thin layer configuration causes the surface concentration of reactants to be determined only by migration effects, in the SERS cell, convection plays a major role in mass transport. Local variations of surface concentration can thus be detected with FT-IR but not with SERS. A feature due to surface cyanide emerges at negative polarisations, starting from -200 mV . This band is located in the range between 2115 and 2090 cm^{-1} and exhibits a relative intensity about ten times higher than the complex CN^- band. This band can be observed up to -800 mV . The same comments as those made for the Au bath apply here. The breadth of the surface band (FWHM ca 37 cm^{-1}) hints at the presence of free cyanide as a minor component of the peak. In the high negative regime, it is possible to observe the partial desorption of surface cyanide, as shown by the evident increase in relative intensity of the free- CN^- peak (numerically very close to the same shift of adsorbed CN^- at -1400 mV), which remains the only one in this frequency range at -1600 mV .

The alloy system shows a complex scenario (Fig. 13), which can be in part interpreted as consisting in features that are typical of the two single-metal systems described above, along with specific peculiarities. Four peaks appear in the region $2000\text{--}2200\text{ cm}^{-1}$ between -600 (ocp) and -1600 mV . Three of these do not show any tuning with potential: 2079 cm^{-1} (free cyanide), 2144 cm^{-1} (Ag(I) cyanocomplex), 2166 cm^{-1} (Au(I) cyanocomplex). A surface feature, with outranging intensity with respect to these bulk species, shifts from 2110 to 2085 cm^{-1} in the same potential region. This surface feature is presumably the sum of several vibrational contributions, because of its large FWHM (ca. 40 cm^{-1}) and asymmetrical shape. In the light of the observations we have made about the surface peaks on Au and Ag, we cannot resolve this complexity simply into differences between cyanide adsorbed on Au or Ag adatoms. More probably, a further factor affecting the CN^- stretching frequency is the nature of adsorption sites developing on the growing electrode, both in the case of simple metals and in the case of alloys. We can thus distinguish between different components of the CN^- stretching peak, because of different sites, on gold and on silver; furthermore, a higher number of components is expected with the alloy, where also the chemical nature of the adsorption site may change.

3.3.2. Low-frequency bands

The investigation of the low-frequency range of the spectral region is expected to provide useful information on the interfacial behaviour peculiar to each system. In fact bonds between metal atoms and light elements vibrate in this region. Inspection of this region of the Raman spectra should be helpful, especially in the case of alloys, to discriminate between the species present at the interface and to identify bonds being formed between the solution species and the electrode. We have recorded high-resolution spectra in the low-frequency region for each of the three systems investigated, in three replicate series.

As far as the single-metal systems are concerned, a common feature is noticed: the presence of two bands in the region between 150 and 450 cm^{-1} , the lower frequency one unshifted by potential tuning, the higher frequency one affected by the Stark effect. On the basis of the potential range of existence of such peaks and of the correlation with bands appearing in the high frequency range, we maintain that the former peak belongs to the M–C bond in the complex present at the interface and that the latter is related to the bond formed by adsorbed CN^- and the metal surface. This interpretation is compatible with the different potential dependence of the band position. The alloy system shows a more complex behaviour.

On gold two bands are present in the potential range between -700 and -1400 mV . The band at ca. 295 cm^{-1}

is ubiquitous and does not show an evident shift with potential. It is present also in the range $(-700, -800\text{ mV})$ where no trace of adsorbed CN^- appears. The band at 360 cm^{-1} appears along with the appearance of the adsorbed CN^- stretching band (at -900 mV and more negative potentials) and displays a Stark-shift of ca. $34\text{ cm}^{-1}\text{ V}^{-1}$. We have thus assigned the band at 290 cm^{-1} to Au(I)–C stretching in the gold cyanocomplex in the double layer and the band at 360 cm^{-1} to Au⁰–C stretching of adsorbed CN^- on metallic gold, which is expected to display a potential-dependent position. The occurrence of gold complex at the interface and the possibility of giving a SERS signal has been commented above in this Section. The presence of the complex CN^- stretching band at 2165 cm^{-1} is observed when the band at 290 cm^{-1} is the only one recorded at low wavenumbers. An abrupt change occurs around the threshold of -1500 mV , below which the previously mentioned bands disappear and two new bands at ca. 270 cm^{-1} and at ca. 470 cm^{-1} can be noticed.

Examining the low frequency range in the spectra recorded on silver, a band at 220 cm^{-1} is noticed between -300 and -1100 mV . This band does not change its position with potential. In the same potential range a band is observed whose position shifts from 306 cm^{-1} at -300 mV to 297 cm^{-1} at -1100 mV ($26\text{ cm}^{-1}\text{ V}^{-1}$). In the whole potential range investigated, at high wavenumbers, both the band due to the silver cyanocomplex (2143 cm^{-1}) and that due to adsorbed cyanide (ca. 2110 cm^{-1} , with a Stark-shift) are present. In analogy to the case of gold, we assign the band at 220 to Ag(I)–C stretching in the silver cyanocomplex involved in the reduction process and the band at ca. 300 cm^{-1} to the stretching of Ag⁰–C. In the series of spectra shown in Fig. 12, the quality of the spectra is affected by the Notch-filter cut-off that impairs a clear interpretation of the low frequency region between -1100 and -1500 mV , but in the replicate series of the same system measured with a higher-resolution grating (not shown for brevity), with lower filter effect, the same peaks at 220 cm^{-1} and ca. $300\text{--}290\text{ cm}^{-1}$ are apparent in the spectra recorded in this potential interval. At -1600 mV , along with the spectral change at high wavenumbers described in Section 3.3.2, two new bands appear in the region considered here: one located at 298 cm^{-1} (different from the one due to Au⁰–C, that would be located at 290 cm^{-1} at this potential, because of the Stark-shift) and one located at 460 cm^{-1} .

The low frequency region in the alloy system shows discernible peaks between -600 and -1300 mV . The spectral interpretation of such peaks is not as straightforward as expected from the simple superposition of the features observed in the simple metal cases. Shifts in typical peak positions can be caused by the different metal substrate. Mixing of different vibrational modes is also possible, since the peak positions are quite close.

The spectra display the following bands: from -600 to -1300 mV, a broad and poorly resolved peak (a) shifting from 240 to 219 cm^{-1} ; from -600 to -900 mV a peak at 298 cm^{-1} (b) with no Stark-tuning; from -1000 to -1300 mV a broad peak (c) shifting from 292 to 283 cm^{-1} and a peak (d) at 360 cm^{-1} . On the basis of peak position and comparison with data from single-metal baths, we believe that peaks (a) and (b) are due to Ag(I)–C and Au(I)–C stretching, respectively. At potentials corresponding to the highest deposition rates investigated, peaks (c) and (d) appear, most probably due to cyanide adsorbed on silver (ca. 290 cm^{-1}) and gold (360 cm^{-1}), respectively.

4. Conclusions

Ag–Au alloys of compositions ranging from 13% to 75% Au can be electrodeposited from solutions containing Ag(I) and Au(I) cyanocomplexes and free cyanide. The electrodeposited alloys are single-phase fcc and the sequence of peak positions conforms to Végard's law. The crystallites sizes are nanometric and anticorrelated with negative polarisation. The morphology of the alloys is markedly dependent on the electroposition potential and varies from macrocrystalline to globular to cauliflower and eventually to dendritic, with increasing negative polarisation. The electrochemical kinetics of the three systems investigated are similar and imply a reduction of M(I) to M(0) through an adsorbed intermediate at low overvoltages and a direct reduction at high overvoltages. The variations in the degree of negative passivation with the negative overvoltage are due to adsorbed ligands, their nature was assessed by in situ SERS.

In the three systems investigated, a SERS peak due to surface cyanide appears when the electrodeposition starts. SERS bands from surface $\text{Au}(\text{CN})_2^-$ and $\text{Ag}(\text{CN})_2^-$ were also observed. The surface CN^- peaks are believed to consist of several components, related to different surface cyanide types. Changes in peak components can be related to changes in the electrodeposition mechanism, coherently with the voltammetric behaviour. The alloy system shows complex SERS spectra which can be in part explained as an addition of spectral features from the separate single-metal systems, but some specific peculiarities seem to be related to synergistic effects. In particular, the band corresponding to the $\text{C}\equiv\text{N}$ stretching of CN^- adsorbed at the metallic surface exhibits a compound nature, which is presumably the sum of several vibrational contributions due to the presence of adsorption sites with different compositions. The differences in the Stark tuning between the systems investigated can be explained by the fact that the bond between CN^- and the cathode is more covalent in the case of pure Au

and it exhibits a more ionic character in the Ag-containing systems.

The analysis of the low-frequency range of the spectral region provided useful information on the electroodic structure during electrodeposition. A potential-independent band was attributed to M(I)–C stretching due to the adsorbed cyanocomplex and a potential-dependent one due to the M(0)–C vibration of CN^- adsorbed on the growing cathode. The spectral details of low-frequency peaks for the alloy system cannot be explained with a simple superposition approach. Again, shifts in typical peak positions can be caused by the different metal substrate and mixing of different vibrational modes can occur.

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